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Research Report

Self-Optimal Clustering using Threshold Function

Course: EE656 - AI, ML, DL and it's Applications

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Abstract

Self-optimal clustering (SOC) is an improved version of Improved Mountain Clustering(IMC) technique developed by Dr. Nishchal Kumar Verma and Abhishek Roy. SOC has been evaluated via cluster compactness within itself and separation with other clusters. The optimizing factor in the threshold function is computed via interpolation and found to be effective in forming better quality clusters as verified by visual assessment and various standard validation indices like the global silhouette index, partition index and separation index. This makes SOC a reliable and efficient tool for applications in pattern recognition, data mining, and machine learning.

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Section 1

Introduction to Clustering

1.1 What is Clustering?

Imagine you go to a candy shop with hundreds of different candies. You want to group similar ones together – like putting all the chocolate candies in one bowl, the sour candies in another, and so on. This process is called clustering – organizing items so that similar ones stay together.

- A **cluster** is just a group of similar things.
- The **goal** is to make sure:
 - Things in the **same group** are **very similar**.
 - Things in **different groups** are **quite different**

Two types of Clustering

Hard Clustering:

- Each item belongs to only one group.
- *Example:* A student is assigned to one classroom only.

Fuzzy Clustering:

- Each item can belong to multiple groups with different levels of association.
- *Example:* A person can like sports (80%), music (60%), and reading (40%). They're part of all groups, but with different strengths.

You can also convert fuzzy clustering to hard by choosing the group with the highest percentage or score.

1.2 A Few Popular Methods

1. Fuzzy C-Means (FCM):

- Items can belong to multiple clusters.
- It's popular and gives good results, but takes time.

2. Probabilistic Clustering:

- Items belong to one cluster only, based on probability.
- Precise, but takes a lot of time to finish.

3. Mountain Clustering:

- Imagine a mountain-like map where peaks show potential group centers.
- Earlier versions missed good clusters by flattening nearby points too quickly.

4. Improved Mountain Clustering (IMC):

- A better version that keeps checking for more good clusters.
- Faster and gives more accurate groups.

1.3 What this Paper Proposes: SOC (Self-Optimal Clustering)

The authors suggest a new method called Self-Optimal Clustering (SOC):

- It automatically finds the best number of groups.
- Uses mathematical tricks (interpolation) to improve accuracy.
- It's better than older methods in both quality and speed.
- Works well on different kinds of data.

Section 2

Overview of Clustering Techniques

2.1 What’s This Section About?

This section gives a quick tour of different clustering techniques that researchers have developed over time. Just like there are different ways to organize your closet (by color, size, or type), there are many ways to group data depending on what you want to find.

2.2 Different Approaches to Clustering

Scientists and engineers have tried many smart ways to group data better:

- Local variations: Group things based on local patterns (like noticing the colors of nearby houses to group neighborhoods).
- Normalized cuts: Split data like cutting a cake into equal parts — but based on similarities.
- Feature analysis: Focus on important features, like grouping students by their favorite subjects.
- Saddle point detection: Look for “valleys” or “peaks” in data, like finding low points in a mountain to know where one area ends and another begins.
- Multiresolution: Zooming in and out to see details — like looking at a map from far away and then zooming into the street level.

- Color-texture regions: Used especially in image analysis, grouping based on color and feel (imagine sorting clothes by both color and fabric).
- Stochastic clustering: A random-based approach to test different ways of grouping and see what works best.

Cool Techniques You Should Know (With Simple Examples)

1) Karger's Method: Think of this like tearing a paper again and again in different ways until you find the cleanest cut.

2) Unsupervised Segmentation: Automatically figuring out how to group things without a teacher — like kids forming teams on their own.

3) Threshold Classifier Systems: Like setting a passing score to decide who gets into a club.

2.3 Clustering techniques that are used a lot

Here are some popular ones:

- Fuzzy C-Means (FCM): Items can belong to multiple groups.
- Mountain Clustering: Uses a map-like method where high points mean strong candidates for clusters.
- Modified Mountain Clustering (MMC): An improved version that avoids wasting time on bad clusters.
- Expectation-Maximization (EM): A step-by-step process of guessing the group, checking it, and adjusting — like improving your guesses in a guessing game.
- K-Means Clustering:
 - Very common and easy.

- *Example*: If you have test scores, you can group students into "High", "Medium", and "Low" performers.
- K-Medoid Clustering: Similar to K-Means, but instead of using average values, it picks real items (actual students or actual candies) to represent a group.

2.4 What's Hyperspace?

- Hyperspace just means using lots of features to represent data.

Example: A person can be described by height, weight, age, hobbies, and favourite colour — all at once.

In many algorithms, data is handled in this complex, multi-feature space

Limitations of Older Methods

- Some methods, like Mountain Clustering, get slower and messier when data has many dimensions (features).
- That's where MMC came in — to make it faster by reducing unnecessary calculations.

2.5 A Bit of Math Magic: Interpolation

The new SOC technique uses a mathematical trick called Lagrange Interpolation to find better thresholds (like cut-off points for forming groups).

Daily life example: Imagine you're filling a jug with water and trying to predict when it will spill. If you record water levels at different times, interpolation helps you estimate the spill point even if you didn't measure it exactly.

How Is SOC Better?

- SOC is an improved version of IMC.
- It finds the best way to group data by adjusting the threshold mathematically, not just guessing.
- It uses Lagrange interpolation, a smart way to make predictions based on known data points, to improve the accuracy of clustering.

Section 3

Self-Optimal Clustering :SOC

3.1 Algorithm

Step 1 Normalization: the data for each dimension of hyperspace so that the data points are bounded by a unit hypercube. The j th instance of the data in x hyperspace is defined as:

$$\mathbf{x}_j = \begin{pmatrix} x_j^1 \\ x_j^2 \\ \vdots \\ x_j^D \end{pmatrix}$$

where D is the total number of dimensions of the hyperspace.

Let $\bar{\mathbf{x}}_j$ be the normalized instance as:

$$\bar{\mathbf{x}}_j = \frac{\mathbf{x}_j - x_{\min}}{x_{\max} - x_{\min}}, \quad \forall j = 1, 2, \dots, n$$

$$\mathbf{x}_{\min} = \left(\min_{j=1}^n x_j^1, \min_{j=1}^n x_j^2, \dots, \min_{j=1}^n x_j^D \right)$$

$$\mathbf{x}_{\max} = \left(\max_{j=1}^n x_j^1, \max_{j=1}^n x_j^2, \dots, \max_{j=1}^n x_j^D \right)$$

n = total number of instances or data points in the data set.

Step 2: Threshold Computation

Determine the threshold value δ_m , which is a positive value defining the neighborhood of the data points for the m -th cluster. δ_m is a heuristic expression multiplied by an optimizing factor β_m for the m -th cluster. Initially, β_m is set to unity when obtaining the m -th cluster for the first time.

The threshold function is computed as:

$$\delta_m = \left(\frac{1}{2n} \sum_{j=1}^n \min \left(\sum_{i=1}^D x_j^i \right) \right) \cdot \beta_m$$

Step 3: Potential Function Computation

Calculate the potential value P_r^m for each data point $\bar{\mathbf{x}}_r$ in the m -th cluster using the mountain function, which depends on the squared Euclidean distance between $\bar{\mathbf{x}}_r$ and all other normalized data points $\bar{\mathbf{x}}_j$. This is given by:

$$d^2(\bar{\mathbf{x}}_r, \bar{\mathbf{x}}_j) = (\bar{\mathbf{x}}_r - \bar{\mathbf{x}}_j)^T Q (\bar{\mathbf{x}}_r - \bar{\mathbf{x}}_j)$$

where Q is the identity matrix (i.e., $Q = I$).

Then, the potential function is computed as:

$$P_r^m = \sum_{j=1}^n \exp \left(-\frac{d^2(\bar{\mathbf{x}}_r, \bar{\mathbf{x}}_j)}{\delta_m^2} \right)$$

is that value of “r” at which the value of P_r^m is found to be the highest.

Step 4: Select the data point corresponding to the highest value among $P_1^m, P_2^m, \dots, P_n^m$ as the m -th cluster center $\bar{c}_m$. This can be represented as:

$$\bar{c}_m = \bar{x}^* \quad \Leftrightarrow \quad P_m^* = \max_{r=1}^n (P_r^m)$$

Step 5: Cluster Assignment

Assign data points to the m -th cluster whose Euclidean distance from the cluster center $\bar{\mathbf{c}}_m$ is less than or equal to the threshold value δ_m . Mathematically, this is expressed as:

$$d^2(\bar{\mathbf{x}}_r, \bar{\mathbf{c}}_m) \leq \delta_m, \quad \forall r = 1, 2, \dots, n$$

Step 6: Eliminate all those data points from the data set which are assigned to the m th cluster.

Step 7: Repeat Steps 2 to 6 for the reduced data set to make successive clusters, equal to the optimum number of clusters M .

Step 8: Distribute the rest of the data points among the formed clusters depending upon their Euclidean distances, i.e., nearness to the respective cluster centers.

Step 9: Calculate the global silhouette value via the silhouette index (GSI) for the obtained clusters. A GSI value close to unity indicates better cluster formation, and this may be possible when silhouette values S_m for $m = 1, 2, 3, \dots, M$ tend to attain unity value. Let t be any cluster formed for which the threshold and silhouette values are δ_t and S_t , respectively.

Step 10: Interpolation of Threshold vs Silhouette Value

Obtain a relation between the threshold value δ_t and the corresponding silhouette value S_t by interpolating the known pairs (δ_m, S_m) for $m = 1, 2, \dots, M$, using Lagrange's interpolation formula.

For a total of M clusters, we have M pairs:

$$(\delta_1, S_1), (\delta_2, S_2), \dots, (\delta_M, S_M)$$

The interpolation polynomial in Lagrange's form is expressed as:

$$S_t = \sum_{m=1}^M S_m \cdot \ell_m(\delta_t)$$

where $\ell_m(\delta_t)$ is the Lagrange basis polynomial defined by:

$$\ell_m(\delta_t) = \prod_{\substack{k=1 \\ k \neq m}}^M \frac{\delta_t - \delta_k}{\delta_m - \delta_k}$$

Using the above expression, we write:

$$S_t = \sum_{m=1}^M S_m \cdot \prod_{\substack{k=1 \\ k \neq m}}^M \frac{\delta_t - \delta_k}{\delta_m - \delta_k}$$

Expanding the expression:

$$\begin{aligned} S_t = & S_1 \cdot \prod_{k=2}^M \frac{\delta_t - \delta_k}{\delta_1 - \delta_k} + S_2 \cdot \prod_{\substack{k=1 \\ k \neq 2}}^M \frac{\delta_t - \delta_k}{\delta_2 - \delta_k} \\ & + S_3 \cdot \prod_{\substack{k=1 \\ k \neq 3}}^M \frac{\delta_t - \delta_k}{\delta_3 - \delta_k} + \dots + S_M \cdot \prod_{k=1}^{M-1} \frac{\delta_t - \delta_k}{\delta_M - \delta_k} \end{aligned}$$

Further expanding this gives a polynomial expression of S_t in terms of δ_t :

$$\begin{aligned} S_t = & S_1 \cdot \frac{(\delta_t - \delta_2)(\delta_t - \delta_3) \dots (\delta_t - \delta_M)}{(\delta_1 - \delta_2)(\delta_1 - \delta_3) \dots (\delta_1 - \delta_M)} \\ & + S_2 \cdot \frac{(\delta_t - \delta_1)(\delta_t - \delta_3) \dots (\delta_t - \delta_M)}{(\delta_2 - \delta_1)(\delta_2 - \delta_3) \dots (\delta_2 - \delta_M)} \\ & + \dots \\ & + S_M \cdot \frac{(\delta_t - \delta_1)(\delta_t - \delta_2) \dots (\delta_t - \delta_{M-1})}{(\delta_M - \delta_1)(\delta_M - \delta_2) \dots (\delta_M - \delta_{M-1})} \end{aligned}$$

This is a polynomial expression of S_t in terms of δ_t , where the maximum possible value of S_t is unity.

Step 11: Substitute $S_t = 1$ in the obtained polynomial and find the roots (δ_t) of the obtained equation.

Step 12: Substitute the obtained roots (δ_t) back to the polynomial

one by one, and select that root for which the value of S_t is found to be closest to unity. The selected root η is the value of threshold function δ_t corresponding to the maximum value of S_t .

Step 13: Update of Optimizing Factor β_m

Divide η by δ_m , for each cluster $m = 1, 2, 3, \dots, M$, to obtain the corresponding optimizing factors β_m . These values are substituted back into the threshold equation (previously defined in Step 2) to compute new values of δ_m when repeating the clustering process. This update helps in maximizing the silhouette value S_m for each cluster.

$$\beta_m = \frac{\eta}{\delta_m}, \quad \forall m = 1, 2, 3, \dots, M$$

Step 14: Using these new β_m values, repeat from Steps 2 to 14 until δ_m gets converged where the GSI value for the formed clusters is maximized. This gives the best possible cluster through this algorithm.

At the end, select the β_m values corresponding to a particular iteration for which the GSI value is found to be the maximum, and repeat the algorithm at the optimum number of clusters M to get the best possible clusters.

On the basis of wide analysis and close observations of the results in all the cases, the number of repetitions of particular steps in the algorithm is chosen as ten.

3.2 Measures of Cluster Quality

Global Silhouette Index (GSI) For a given cluster X_m , with $m = 1, 2, 3, \dots, M$, the Global Silhouette Index (GSI) assigns to each sample in X_m a quality measure $s(i)$, known as the silhouette width, where $i = 1, 2, 3, \dots, N_m$, and N_m is the number of samples in the m -th cluster.

The silhouette width is a confidence indicator of the membership of the i -th sample in cluster X_m . It is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max \{a(i), b(i)\}}$$

where: - $a(i)$ is the average distance between the i -th sample and all other samples in X_m , - $b(i)$ is the minimum average distance between the i -th sample and all samples in any other cluster X_k (with $k \neq m$), - The value of $s(i)$ lies in the range $-1 \leq s(i) \leq 1$.

A value of $s(i)$ close to 1 indicates that the sample is well-clustered. A value near 0 suggests that the sample could equally belong to a neighboring cluster. Negative values indicate possible misclassification.

The silhouette value for the m -th cluster is the average of the silhouette widths of its members:

$$S_m = \frac{1}{N_m} \sum_{i=1}^{N_m} s(i)$$
$$\text{GSI} = \frac{1}{M} \sum_{m=1}^M S_m$$

In many cases, GSI can be applied to estimate the most appropriate number of clusters (M). In those cases, the partition with the maximum GSI is taken as the optimal partition.

Partition Index (PI) The Partition Index (PI) is defined as the ratio of the compactness and separation of the clusters. It is computed as the sum of

individual cluster validity measures, normalized by the fuzzy cardinality of each cluster. Mathematically:

$$\text{PI} = \frac{\sum_{m=1}^M \frac{\sum_{j=1}^n \mu_{jm}^2 \|\mathbf{x}_j - \mathbf{c}_m\|^2}{N_m}}{\sum_{k=1}^M \|\mathbf{c}_k - \mathbf{c}_m\|^2}$$

where: - $\mathbf{c}_m$ is the center of the m -th cluster, - $\mu_{jm} \in \{0, 1\}$ is the membership of data point $\mathbf{x}_j$ in cluster m , - $N_m = \sum_{j=1}^n \mu_{jm}$ is the fuzzy cardinality of cluster m , - $\|\cdot\|$ denotes the Euclidean norm.

The constraints on memberships are:

$$\begin{aligned} \sum_{m=1}^M \mu_{jm} &= 1, \quad \forall j = 1, 2, \dots, n \\ 0 &\leq \sum_{j=1}^n \mu_{jm} \leq N, \quad \forall m = 1, 2, \dots, M \end{aligned}$$

A lower value of PI indicates a better partitioning of the dataset.

Separation Index (SI) The Separation Index (SI) is based on minimum-distance separation and measures partition validity using inter-cluster distances. It is defined as:

$$\text{SI} = \frac{\sum_{m=1}^M \sum_{j=1}^n \mu_{jm}^2 \|\mathbf{x}_j - \mathbf{c}_m\|^2}{n \cdot \min_{\substack{k, m \\ k \neq m}} \|\mathbf{c}_k - \mathbf{c}_m\|^2}$$

A lower value of SI also indicates better clustering, with well-separated and compact clusters.

Need for Optimum Threshold Function There is a need to optimize the threshold function because of a good number of existing cases

in which IMC-1 or IMC-2 remains unable to show better quality results in comparison to other clustering techniques. Also, with the threshold function of IMC-2 being modified on heuristic basis, there remains a lot of scope in the estimation of the heuristic factor in its threshold function for further improvisation. The threshold function in SOC is optimized using interpolation method, in order to obtain the best possible clustering with this method.

The improvement in the quality of clustering with SOC is well supported with the relative increment in the GSI value and the significant decrement in the PI and SI values.

Determining Optimum Number of Clusters : To determine the optimum number of clusters, calculate GSI obtained via IMC technique for various clusters. With the obtained GSI values, that M is said to be optimum whose corresponding GSI value is found to be maximum.

Section 4

Results

4.1 Running the SOC algorithm on Test Image

The code implementation of SOC was run on a Test Image and below are the results.

Higher GSI implies better cluster, Lower SI implies better the cluster.



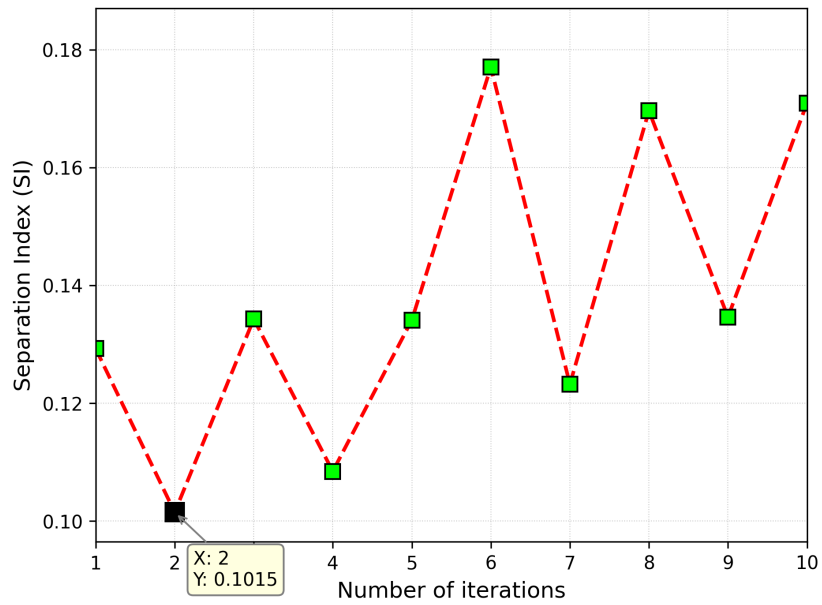
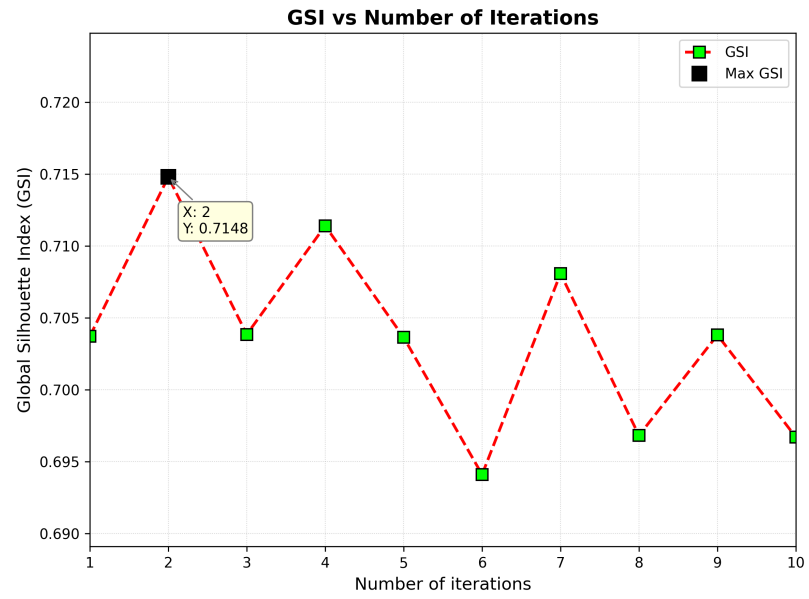
Test Image before clustering



Image after clustering

Iter	δ_1	δ_2	δ_3	δ_4	S_1	S_2	S_3	S_4	GSI	η	β_1	β_2	β_3	β_4
1	0.0695	0.1014	0.1087	0.1289	0.8587	0.6425	0.5433	0.7705	0.7037	0.1337	1.0000	1.0000	1.0000	1.0000
2	0.1337	0.1354	0.1437	0.1461	0.8198	0.6498	0.5454	0.8442	0.7148	0.1321	1.9224	1.3188	1.2301	1.0368
3	0.0687	0.0989	0.0998	0.1152	0.8593	0.6407	0.5496	0.7658	0.7038	0.1163	0.9882	0.9757	0.9189	0.9042
4	0.1177	0.1201	0.1334	0.1393	0.8239	0.6455	0.5653	0.8108	0.7114	0.1160	1.6934	1.1764	1.1662	1.0101
5	0.0685	0.0979	0.0944	0.1058	0.8594	0.6409	0.5482	0.7661	0.7036	0.0671	0.9853	0.9662	0.8696	0.8330
6	0.0681	0.0694	0.0755	0.0767	0.8250	0.7265	0.4495	0.7754	0.6941	0.0773	0.9792	0.6849	0.7102	0.6339
7	0.0789	0.1127	0.1134	0.1330	0.8572	0.6294	0.5583	0.7875	0.7081	0.0795	1.1350	1.1133	1.0241	1.0073
8	0.0701	0.0715	0.0748	0.0724	0.8288	0.7159	0.4704	0.7722	0.6968	0.0695	1.0075	0.7055	0.7009	0.5978
9	0.0690	0.0986	0.1009	0.1223	0.8584	0.6424	0.5487	0.7658	0.7038	0.0711	0.9926	0.9724	0.9294	0.9598
10	0.0716	0.0731	0.0752	0.0704	0.8262	0.7198	0.4693	0.7715	0.6967	0.0796	1.0299	0.7211	0.7043	0.5811

Figure 4.1: Parameters over various iterations



Optimum no. of Clusters = 4

Clustering Method	GSI	SI
IMC-1	0.7037	0.1293
IMC-max	NaN	0.2665
IMC-half	NaN	0.531
IMC-2	0.6984	0.1566
SOC	0.7148	0.1015

NaN DENOTES 'NOT A NUMBER'

Figure 4.2: Comparison of various clustering techniques

Section 5

Conclusion

SOC is parameter free, density-based clustering algorithm that intelligently determines both the number of clusters and the optimal threshold for cluster formation.

Produces high-quality, well-separated clusters across a variety of test scenarios.

SOC applications are in Pattern recognition, data mining, bioinformatics and sensor networks etc.

Its performance, validated through both statistical and visual tools, positions it as a strong alternative to conventional clustering methods, especially in scenarios where prior knowledge of data structure is unavailable.